



Iteration (Looping) in C++

- In computer programming, loops are used to repeat a block of code
- A loop statement allows us to execute a statement or group of statements multiple times and following is the general form of a loop statement in most of the programming languages.
- For example, let's say we want to show a message 100 times. Then instead of writing the print statement 100 times, we can use a loop.

There are 3 types of loops in C++

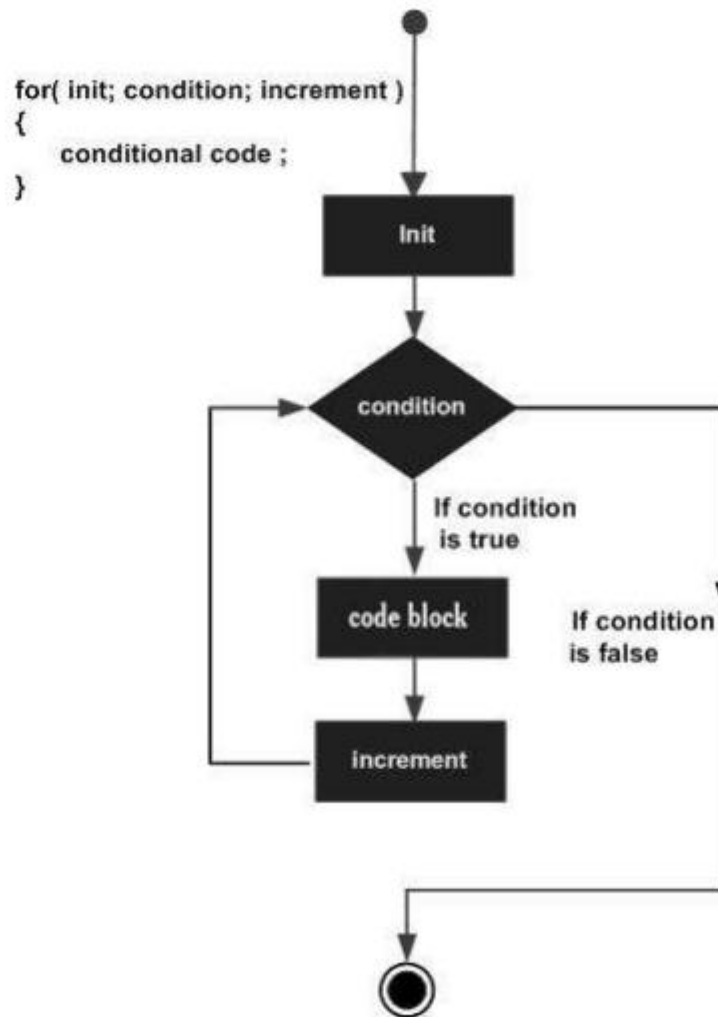
- 1) for loop
- 2) while loop
- 3) do...while loop

1. for Loop

- A for loop is a repetition control structure that allows you to efficiently write a loop that needs to execute a specific number of times.
- Syntax :

```
for ( init; condition; increment )  
{  
    statement(s);  
}
```

Flow Diagram



Questions:

Q. Write a program to print your name 10 times.

Q1. Write a program to print numbers from 1 to 100 using a for loop.

Q2. Write a program to print all even numbers b/w 1 to 50 using a for loop.

Q3. Write a program to calculate and display the sum of first N natural numbers using a for loop.

Q4. Write a program to print the multiplication table of a given number using a for loop.

Q5. Write a program to find the factorial of a number using a for loop.

Q6. Write a program to count how many digits are present in a given number using a for loop.

Q7. Write a program to reverse a given number using a for loop.

Q8. Write a program to print the Fibonacci series up to N terms using a for loop.

What is Fibonacci Series → The Fibonacci series is a sequence of number where each number is the sum of the two previous numbers.

The Fibonacci Sequence

1,1,2,3,5,8,13,21,34,55,89,144,233,377...

$$1+1=2$$

$$1+2=3$$

$$2+3=5$$

$$3+5=8$$

$$5+8=13$$

$$8+13=21$$

$$13+21=34$$

$$21+34=55$$

$$34+55=89$$

$$55+89=144$$

$$89+144=233$$

$$144+233=377$$

Q9. Write a program to check whether a number is prime or not using a for loop.

Prime Numbers to 100

A prime number can only be divided (without a remainder) by itself and 1.

2	3	5	7	11
13	17	19	23	29
31	37	41	43	47
53	59	61	67	71
73	79	83	89	97

sciencenotes.org

Q10. Write a program to find the sum of digits of a number using a for loop.

Q11. write a program to check whether a number is Armstrong number or not.

Armstrong = The sum of each digit raised to the power of the total number of digit is equal to the original number.

$$153 = 1^3 + 5^3 + 3^3 = 1 + 125 + 27 = 153$$

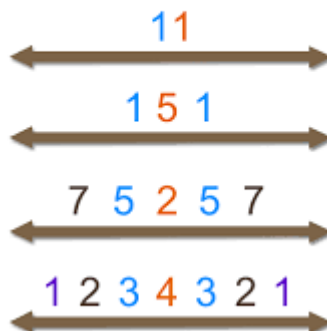
$$371 = 3^3 + 7^3 + 1^3 = 27 + 343 + 1 = 371$$

$$9474 = 9^4 + 4^4 + 7^4 + 4^4 = 6561 + 256 + 2401 + 256 = 9474$$

Q12. Write a program to check whether a number is palindrome or not using for loop.

Palindrome Numbers

Palindrome numbers remain the same whether written forwards or backwards



© w3resource.com

All Pyramid Patterns

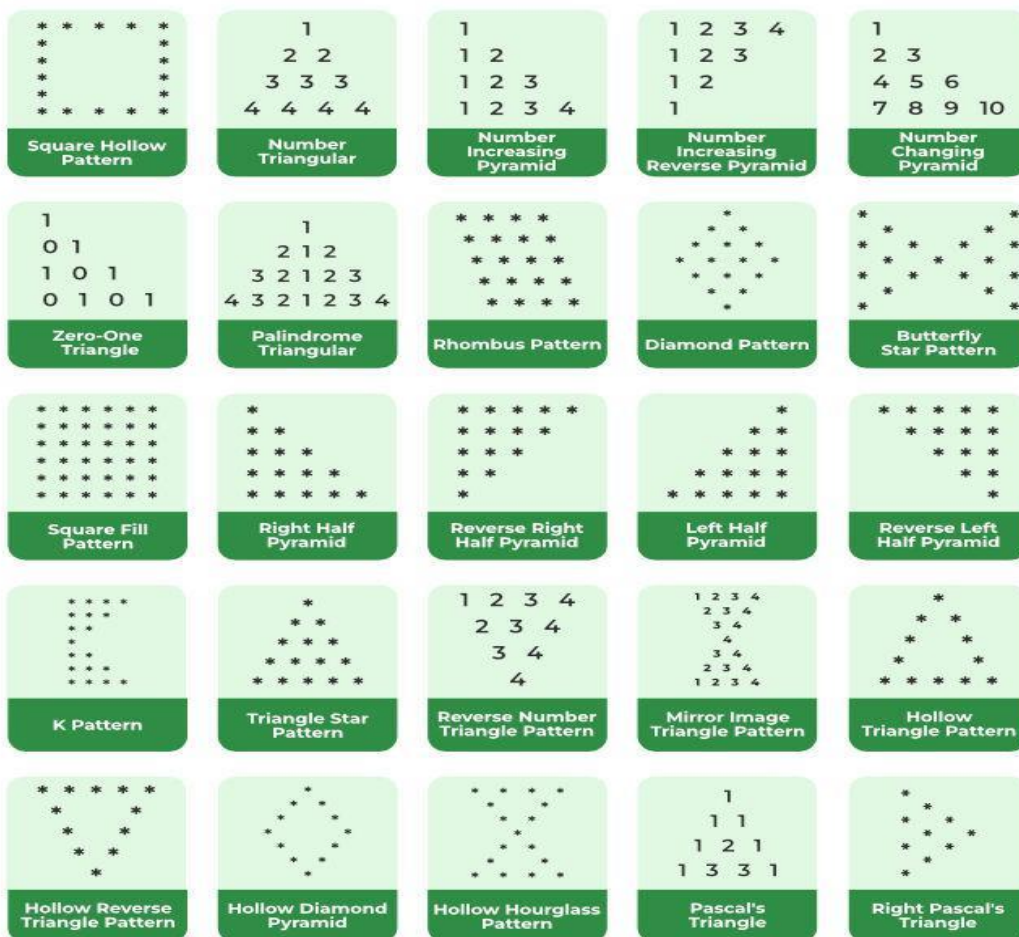
```
1
2 3
4 5 6
7 8 9 10
```

```
1
2 2
3 3 3
4 4 4 4
```

```
*
* * *
* * * * *
* * * * * * *
```

```
* * * * *
* * * *
* * *
*
```

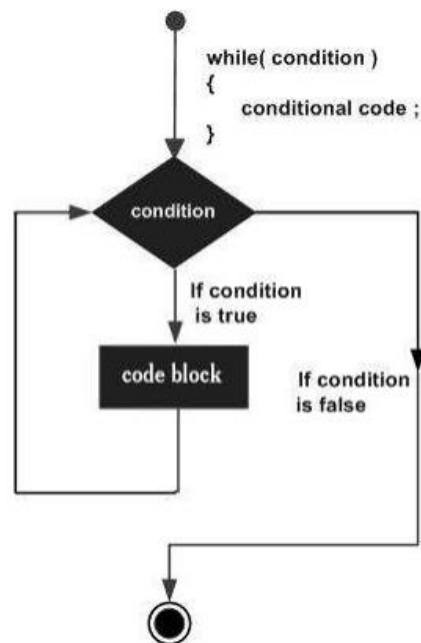
```
A
A B
A B C
A B C D
```



2. While Loop

- A while loop statement repeatedly executes a target statement as long as a given condition is true.
- Syntax

```
while(condition)
{
    statement(s);
}
```



Questions

- Q1.** Write a program to print numbers from 1 to 100 using a while loop.
- Q2.** Write a program to print all odd numbers between 1 and 50 using a while loop.
- Q3.** Write a program to find the sum of first N natural numbers using a while loop.
- Q4.** Write a program to display the multiplication table of a given number using a while loop.
- Q5.** Write a program to calculate the factorial of a number using a while loop.
- Q6.** Write a program to count the digits of a given number using a while loop.
- Q7.** Write a program to reverse a number using a while loop.

Q8. Write a program to check whether a number is prime or not using a while loop.

Q9. Write a program to print Fibonacci series up to N terms using a while loop.

Q10. Write a program to find the sum of digits of a number using a while loop.

3. do...while Loop

- Unlike for and while loops, which test the loop condition at the top of the loop, the do...while loop checks its condition at the bottom of the loop.
- A do...while loop is similar to a while loop, except that a do...while loop is guaranteed to execute at least one time.

- Syntax

```
do
{
    statement(s);
}
while( condition );
```

Questions

1. Write a program to print numbers form 1 to 10 using a do-while loop.
2. Write a program to print numbers form 10 to 1 in reverse order using a do-while loop.
3. Write a program to print all even numbers b/w 1 to 50 using a do-while loop.
4. Write a program to print the multiplication table of a given number using a do-while loop.
5. Write a program to find the sum of first n natural numbers using a do-while loop.
6. Write a program to count the number of digits in a given number using a do-while loop.
7. Write a program to find the reverse of a number using a do-while loop.

8. Write a program to check whether a number is palindrome or not using a do-while loop.
9. Write a program to calculate factorial of a number using a do-while loop.
10. Write a program to display a menu repeatedly until the user choose to exit (using do-while loop)